

Udit Agarwal

+1-930-904-4901 | agarwaludit13@gmail.com | linkedin.com/in/udit013 | github.com/Udit013 | uditagarwal.vercel.app

EDUCATION

Indiana University Bloomington

Master of Science in Computer Science, GPA: 3.84/4.0

Bloomington, Indiana, US

Aug 2024 - May 2026

Kalinga Institute of Industrial Technology (KIIT University)

Bachelor of Technology in Computer Science and Engineering, GPA: 8.85/10.0

Bhubaneswar, Odisha, India

Aug 2020 - May 2024

EXPERIENCE

Heartland Community Network

Senior Consultant

Jun 2026 - Present

Remote, Indiana, US

- Design and develop full-stack applications, backend services, and REST APIs tailored to client operational requirements, contributing across the complete development lifecycle from design through production deployment
- Build AI-powered solutions leveraging LLMs, data analytics, and workflow automation to streamline business processes and improve decision-making for clients across multiple industries

Global Health Impact Project

Software Engineer

Jan 2026 - Jun 2026

Indiana University, Bloomington, Indiana

- Built full-stack features and REST APIs for a global health analytics platform (React, TypeScript, Python), enabling researchers to evaluate pharmaceutical intervention effectiveness across populations
- Developed scalable backend services and data-processing pipelines supporting treatment-coverage modeling, efficacy analysis, and large-scale forecasting workflows

University Information Technology Services (UITS)

Consultant - IT Services

Aug 2025 - Present

Indiana University, Bloomington, Indiana

- Resolved 100+ technical issues weekly across desktop, mobile, and enterprise systems using structured troubleshooting and root-cause analysis
- Supported identity and access management workflows, ensuring secure authentication and account lifecycle management across university systems

PROJECTS

Brain Tumor MRI Classifier - Production Serving (IEEE 2024) | Python, ONNX Runtime, FastAPI,

Docker, TensorFlow/Keras, Grad-CAM

- Served a calibrated MRI classifier as an ONNX + FastAPI endpoint returning class, confidence, and Grad-CAM overlays at 33 ms CPU inference, containerized with Docker
- Co-authored the underlying IEEE 2024 publication benchmarking four CNN architectures on 7,000+ scans, with EfficientNetB3 best at just 11.7M parameters
- Audited the dataset with perceptual-hash deduplication to remove train/test leakage and validated honest performance (93.9% accuracy, 0.985 macro ROC-AUC)

Automated Secrets Scanner | Python, FastAPI, React, TypeScript, PostgreSQL, WebSocket

- Engineered a secret-detection engine covering 26 credential types using regex matching, Shannon-entropy analysis, and false-positive suppression, reaching 95% precision, 94% recall, and 0.94 F1
- Built an asynchronous FastAPI backend with 18 REST endpoints, WebSocket scan progress, recurring scheduling, and SMTP alerting, plus differential scanning to track new vs resolved secrets across runs
- Implemented SARIF 2.1.0, JSON, and CSV export for GitHub Advanced Security and CI/CD integration, with git-history scanning and a React security dashboard for severity analytics

AI Voice Interview Simulator | Next.js, TypeScript, Firebase, Web Speech API, Gemini, Zod

- Built a full-stack platform conducting end-to-end voice interviews with browser-native speech recognition and synthesis, eliminating paid STT/TTS dependencies
- Engineered a stateful conversational engine on Gemini that generates role-specific questions and adapts follow-ups using full-session transcript context
- Developed a Zod-validated evaluation pipeline scoring candidates across five competency dimensions, with persistent history via Firebase Auth and Firestore

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

Frontend: React, Next.js, Tailwind CSS

Backend: FastAPI, Fastify, Node.js, Express.js, REST APIs, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite

AI and ML: PyTorch, TensorFlow, scikit-learn, Hugging Face, LLMs (RAG, Fine-Tuning), ONNX

Cloud and DevOps: AWS, Docker, CI/CD, Git, Postman